MATH 345 Skeleton Notes

Unit 3: Beyond linear maps: local linearization and first-order optimization

Section 3.4: Local linearization and differentiability

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Tangent planes and scalar local linearization

Linear approximations and differentiability

Recall that the linear approximation of a function $f : \mathbb{R} \to \mathbb{R}$ of one variable at some point a is the affine function $L(x) = f(a) + f'(a)(x - a)$, and this function gives the best local affine approximation of f near a. Similarly, the partial derivatives of a function can be used to define a local approximation to a function of several variables. This leads to Linear Approximation.

Definition: Linear Approximation

Given a scalar-valued function f, and a point $\mathbf{a}$ in the domain of f where the partial derivatives of f are well-defined,

the linear approximation of f at the point $\mathbf{a}$ is

the affine function

$$L(\mathbf{x}) = f(\mathbf{a}) + (\nabla f)(\mathbf{a})^T(\mathbf{x} - \mathbf{a}),$$

where we can expand the dot product that occurs in the equation, so that

$$L(\mathbf{x}) = f(\mathbf{a}) + \sum_{i=1}^{n} \frac{\partial f}{\partial x_i}(\mathbf{a})(x_i - a_i).$$

The approximation is affine in $\mathbf{x}$; its linear part is the map $\mathbf{h} \mapsto (\nabla f)(\mathbf{a})^T \mathbf{h}$ on small input changes.

With the base point $\mathbf{a}$ fixed, the gradient $\nabla f(\mathbf{a})$ is a fixed vector. By Linear functions are dot products, its dot product with $\mathbf{h}$ defines a linear function of the input change $\mathbf{h}$:

$$\ell_{\mathbf{a}}(\mathbf{h}) = \nabla f(\mathbf{a}) \cdot \mathbf{h}.$$

This is the linear part of the approximation. Taking $\mathbf{h} = \mathbf{x} - \mathbf{a}$, we can write

$$L(\mathbf{x}) = f(\mathbf{a}) + \ell_{\mathbf{a}}(\mathbf{x} - \mathbf{a}).$$

In particular, the linear approximation of a scalar-valued function $f(x, y)$ at a point (a, b) is given by

$$L(x, y) = f(a, b) + \frac{\partial f}{\partial x}(a, b)(x - a) + \frac{\partial f}{\partial y}(a, b)(y - b).$$

Activity: Linear approximation in three variables (U3-LO4)

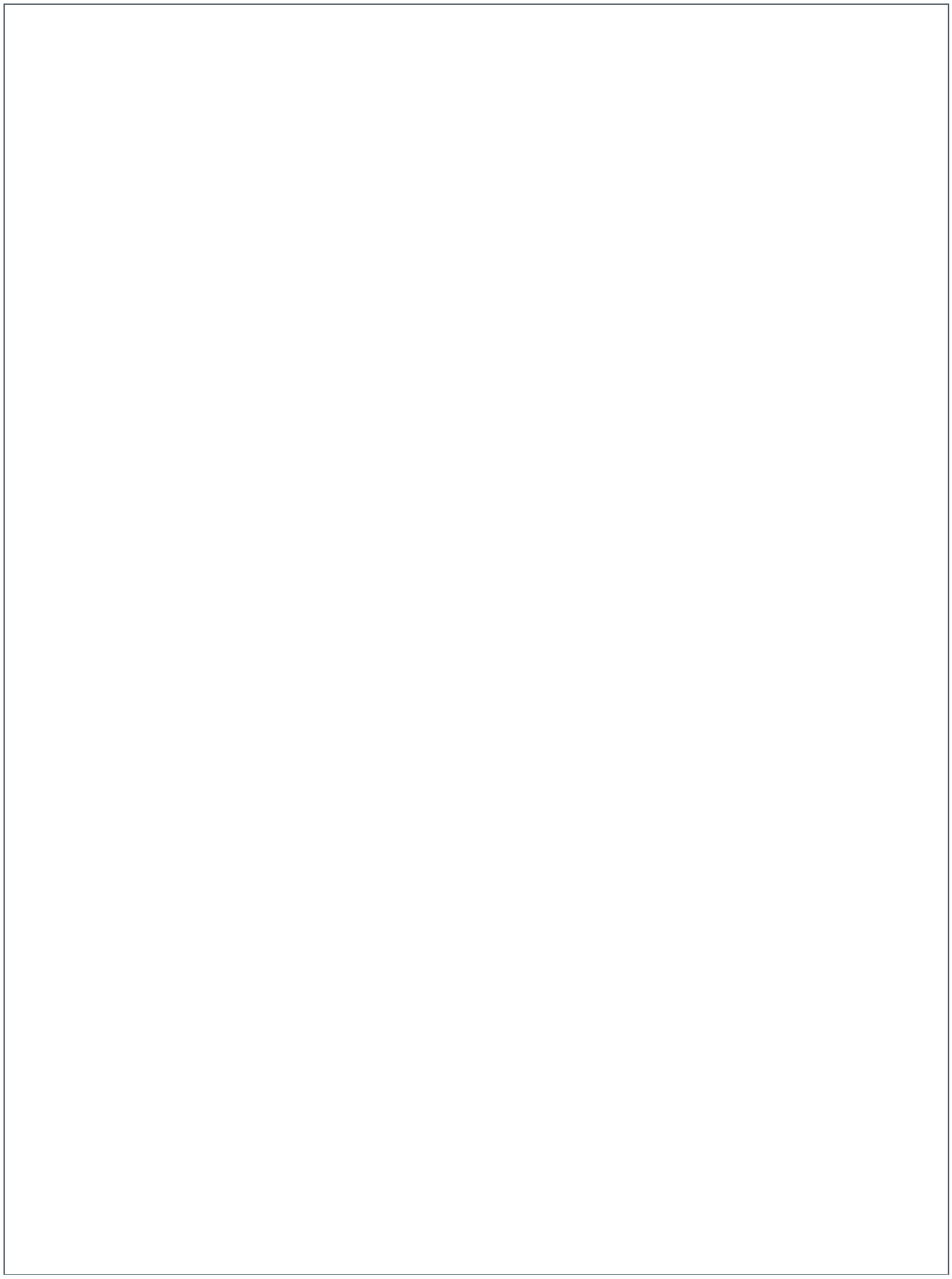
Consider the function

$$f(x, y, z) = x^2 y + y z^2.$$

Find the linear approximation of the function

$$f(x, y, z) = x^2 y + y z^2$$

at the point $(1, 2, 0)$.



Activity: Linear approximation in three variables (U3-LO4)

Consider the function

$$f(x, y, z) = x^2y + yz^2.$$

Use the linear approximation you have found to approximate the value $f(1.04, 1.97, 0.05)$.

Activity: Predicting a change in fitting error (U3-LO4)

For the prediction loss in From predictions to a scalar error function (U3-LO1, U2-LO1), How model weights change the loss (U3-LO3) gives

$$\mathscr{L}(\mathbf{w}_0) = 1, \qquad \nabla\mathscr{L}(\mathbf{w}_0) = \begin{bmatrix} 0 \\ -2 \end{bmatrix}, \qquad \mathbf{w}_0 = (1, 1).$$

Suppose we increase both weights by $1/10$, so $\mathbf{h} = (1/10, 1/10)$.

1. Find the affine approximation $\widehat{\mathscr{L}}(\mathbf{w})$ near $\mathbf{w}_0$. Use it to predict the change in loss and the new loss at $\mathbf{w}_0 + \mathbf{h}$.
2. Compute the new predictions $X(\mathbf{w}_0 + \mathbf{h})$, residuals, and exact loss. Compare the exact loss with the affine prediction.
3. For an arbitrary change $\mathbf{h} = (h_1, h_2)$, expand $\mathscr{L}(\mathbf{w}_0 + \mathbf{h})$. Show that the approximation error is $\|X\mathbf{h}\|^2$. What happens to this error if the change is halved?

Now suppose a function f of a single variable is differentiable at a point a, i.e., so that the limit

$$f'(a) = \lim_{x \to a} \frac{f(x) - f(a)}{x - a}$$

exists. We can rearrange this equation to read

$$\lim_{x \to a} \frac{f(x) - f(a) - f'(a)(x - a)}{x - a} = 0,$$

i.e., so that

$$\lim_{x \to a} \frac{f(x) - L(x)}{x - a} = 0,$$

where $L(x) = f(a) + f'(a)(x - a)$ is the linear approximation of f at a. A function of several variables is *differentiable* when the same property holds, with the linear approximation given by the partial derivatives of the function.

Definition: Differentiability

A scalar-valued function f is differentiable at a point $\mathbf{a}$ if the gradient of f is well-defined at $\mathbf{a}$, and if

$$\lim_{\mathbf{x} \to \mathbf{a}} \frac{f(\mathbf{x}) - L(\mathbf{x})}{\|\mathbf{x} - \mathbf{a}\|} = 0,$$

where

$$L(\mathbf{x}) = f(\mathbf{a}) + (\nabla f)(\mathbf{a})^T(\mathbf{x} - \mathbf{a})$$

is the linear approximation of f at $\mathbf{a}$. A vector-valued function is differentiable at a point precisely when each of its component functions is differentiable at that point.

Note

The equation

$$\lim_{\mathbf{x} \to \mathbf{a}} \frac{f(\mathbf{x}) - L(\mathbf{x})}{\|\mathbf{x} - \mathbf{a}\|} = 0$$

can only hold if L is a *good* linear approximation of f near $\mathbf{a}$, so that the magnitude of the numerator is much smaller than the magnitude of the denominator when $\mathbf{x}$ is near $\mathbf{a}$, i.e., when $\|\mathbf{x} - \mathbf{a}\|$ is small.

The following theorem is the main way we will determine if a function is differentiable.

Theorem: Continuity of First Partials Implies Differentiability

Let f be a scalar-valued function, and suppose $\mathbf{a}$ is an interior point (recall Interior, boundary, and closed sets) of the domains of f and its partial derivatives. Moreover, suppose that all of the partial derivatives $D_1 f, \ldots, D_n f$ are continuous at $\mathbf{a}$. Then f is differentiable at $\mathbf{a}$.

Tangent planes

Activity: A plane containing two lines (U2-LO1)

Find an equation for the plane containing the lines L_1 and L_2, where L_1 is the line parameterized by the function

$$\mathbf{r}(t) = \begin{bmatrix} 1 \\ 1 \\ -2 \end{bmatrix} + t \begin{bmatrix} 2 \\ 3 \\ 1 \end{bmatrix}$$

and where L_2 is the line parameterized by the function

$$\mathbf{r}(s) = \begin{bmatrix} 1 \\ 1 \\ -2 \end{bmatrix} + s \begin{bmatrix} -3 \\ 1 \\ 4 \end{bmatrix}.$$

Definition: Tangent Plane

Let S be a surface, and $\mathbf{x}_0$ a point on S. A **tangent plane** to S at $\mathbf{x}_0$ is a plane that contains the tangent lines of any differentiable curve contained in S as it passes through $\mathbf{x}_0$.

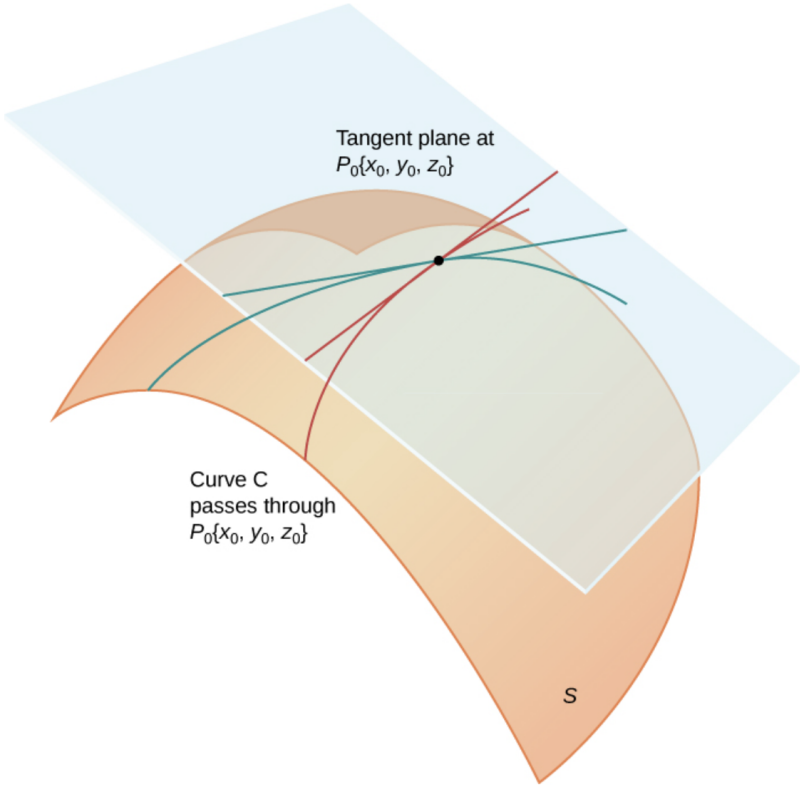


Figure. A tangent plane touching a surface along tangent lines through a point.

Theorem: Equation of the Tangent Plane

Suppose S is the graph of a function $f(x, y)$ which is differentiable at a point (x_0, y_0). Then the tangent plane to S exists, is unique, and is described by the equation

$$z - f(x_0, y_0) = f_x(x_0, y_0)(x - x_0) + f_y(x_0, y_0)(y - y_0).$$

Note that this plane is the graph of the linear approximation L to f at (x_0, y_0).

Why is this true?

We verify that the plane described by the equation above is the only

possible tangent plane. Recall from Geometric Interpretation that the curve obtained by the intersection of S with the plane $y = y_0$ has slope $f_x(x_0, y_0)$ at the point (x_0, y_0), and the tangent line to the curve at this point is by a line with direction vector $(1, 0, f_x(x_0, y_0))$. Similarly, the curve obtained by the intersection of S with the plane $x = x_0$ has slope $f_y(x_0, y_0)$, and so the tangent line to the curve at this point is a line with direction vector $(0, 1, f_y(x_0, y_0))$. But now using the method of A plane containing two lines (U2-LO1), we see that a normal vector to the plane containing these two lines is of the form $\mathbf{n} = (-f_x(x_0, y_0), -f_y(x_0, y_0), 1)$. So the tangent plane to S at $(x_0, y_0, f(x_0, y_0))$ must be described by the equation

$$(-f_x(x_0, y_0), -f_y(x_0, y_0), 1) \cdot (x - x_0, y - y_0, z - f(x_0, y_0)) = 0,$$

an equation we may rearrange to read

$$z - f(x_0, y_0) = f_x(x_0, y_0)(x - x_0) + f_y(x_0, y_0)(y - y_0).$$

Activity: A tangent plane to a surface (U3-LO4)

Find an equation of the tangent plane to the surface

$$z = e^x \cos y$$

at $(0, \frac{\pi}{2}, 0)$.

An important property of the gradient of a function at a point is that it is a vector that is perpendicular to the level curves of the function. We can now use tangent planes to explain this property.

Theorem

Let $f(x, y)$ be a function with continuous partial derivatives. If (a, b) is a point where $\nabla f(a, b) \neq \mathbf{0}$, then $\nabla f(a, b)$ is

perpendicular to the level curve of f passing through the point (a, b).

Why is this true?

By Continuity of First Partials Implies Differentiability, the continuous partial derivatives make f differentiable at (a, b). Let $c = f(a, b)$. By Equation of the Tangent Plane, the tangent plane to the graph of f at (a, b, c) has equation

$$z - c = \frac{\partial f}{\partial x}(a, b)(x - a) + \frac{\partial f}{\partial y}(a, b)(y - b).$$

Along the level curve $f(x, y) = c$, the height is constant. Lifting this curve onto the graph of f therefore places it entirely in the horizontal plane $z = c$. If $\mathbf{v} = (v_1, v_2)$ is a nonzero tangent direction to the level curve at (a, b), then the lifted curve has tangent direction $(v_1, v_2, 0)$: there is no vertical change.

The tangent line to this lifted curve lies in the tangent plane. Substituting a point $(a + tv_1, b + tv_2, c)$ on that tangent line into the plane equation gives

$$0 = t\left(\frac{\partial f}{\partial x}(a, b)v_1 + \frac{\partial f}{\partial y}(a, b)v_2\right).$$

For $t \neq 0$, dividing by t yields

$$\nabla f(a, b) \cdot \mathbf{v} = \frac{\partial f}{\partial x}(a, b)v_1 + \frac{\partial f}{\partial y}(a, b)v_2 = 0.$$

A zero dot product means that the gradient is perpendicular to the tangent direction of the level curve.

See the figure for a graphical example of the theorem.

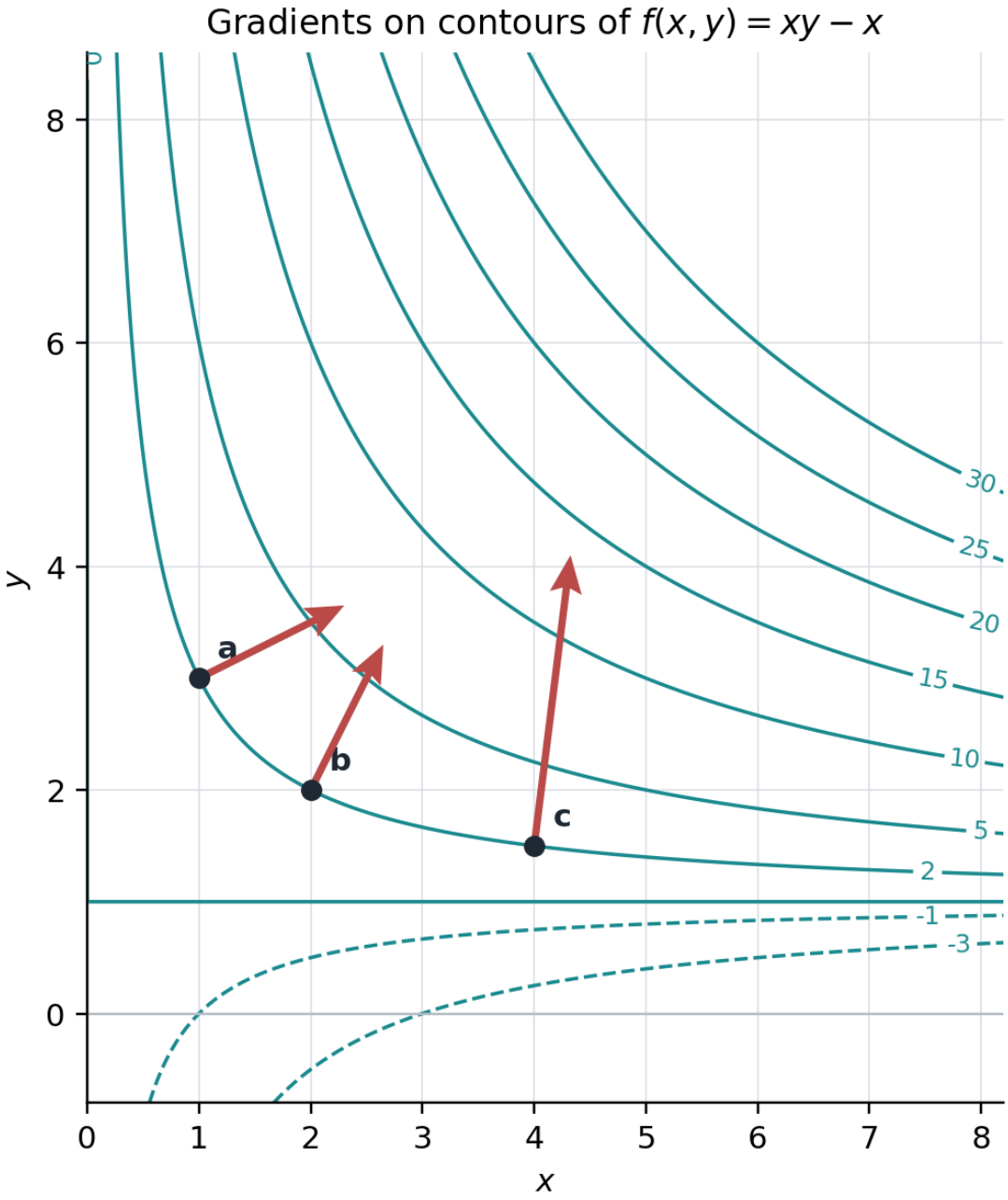


Figure. A contour plot of the function $f(x, y) = xy - x$, with the gradient vectors at three points $\mathbf{a} = (1, 3)$, $\mathbf{b} = (2, 2)$, and $\mathbf{c} = (4, 3/2)$. Adapted from: Stanford's MATH 51 textbook

Vector-valued local linearization

Linear maps reappear locally

A nonlinear map is usually not linear globally. Near one input, however, differentiability replaces small input changes by a linear map.

$$F(\mathbf{a}+\mathbf{h}) \approx F(\mathbf{a}) + J_F(\mathbf{a})\mathbf{h}.$$

$\mathbf{x} \mapsto F(\mathbf{a}) + J_F(\mathbf{a})(\mathbf{x}-\mathbf{a})$ is affine in $\mathbf{x}$. $\mathbf{h} \mapsto J_F(\mathbf{a})\mathbf{h}$ is the linear map on input changes.
Linear algebra gave us global linear maps $\mathbf{x} \mapsto A\mathbf{x}$. Calculus gives us local linear maps $\mathbf{h} \mapsto J_F(\mathbf{a})\mathbf{h}$. The same matrix language therefore reappears inside nonlinear problems.

Definition: Linear Approximation of Vector-Valued Functions

Let $F : \mathbb{R}^n \to \mathbb{R}^m$ be a vector-valued function. Provided that the partial derivatives of the components of F all exist at $\mathbf{a}$, so that the Jacobian matrix J_F is well-defined,

the linear approximation of F at $\mathbf{a}$ is the vector-valued affine function

$$L(\mathbf{x}) = F(\mathbf{a}) + J_F(\mathbf{a})(\mathbf{x}-\mathbf{a})$$

where $J_F(\mathbf{a})(\mathbf{x}-\mathbf{a})$ is the Jacobian matrix of F evaluated at $\mathbf{a}$, and then applied to the vector $\mathbf{x}-\mathbf{a}$. The approximation is affine in $\mathbf{x}$; its linear part is the map $\mathbf{h} \mapsto J_F(\mathbf{a})\mathbf{h}$ on small input changes.
The components of L are precisely the linear approximations of the components of F, i.e., $L_i(\mathbf{x}) = F_i(\mathbf{a}) + (\nabla F_i)(\mathbf{a})^T(\mathbf{x}-\mathbf{a})$.

Note

Geometrically, the graph of the linear approximation to a vector-valued function can be visualized as a higher dimensional "tangent plane" of the graph of the function F at $\mathbf{a}$.

Activity: Approximating a two-component map (U3-LO5)

Let $F(x, y) = (x^2 y, xe^y)$. Find the linear approximation of F at the point $(1, 0)$, and use it to approximate $F(1.1, -0.2)$.

Activity: Local square-grid visualization (U3-LO5)

Return to

$$F\left(\begin{bmatrix} x \\ y \end{bmatrix}\right) = \begin{bmatrix} x + y^2 \\ y \end{bmatrix}.$$

The Jacobian matrix is

Activity: Local square-grid visualization (U3-LO5) (continued)

$$J_F(x, y) = \begin{bmatrix} 1 & 2y \\ 0 & 1 \end{bmatrix}.$$

Let

$$\mathbf{a} = \begin{bmatrix} 0 \\ 1/2 \end{bmatrix}.$$

1. Compute $J_F(\mathbf{a})$.
2. Let $\mathbf{h} = \begin{bmatrix} h_1 \\ h_2 \end{bmatrix}$. Compute $F(\mathbf{a} + \mathbf{h}) - F(\mathbf{a})$.
3. Compute $J_F(\mathbf{a})\mathbf{h}$.
4. Compute the error

$$F(\mathbf{a} + \mathbf{h}) - F(\mathbf{a}) - J_F(\mathbf{a})\mathbf{h}.$$

5. If $0 \leq h_2 \leq r$, how large can the first coordinate of the error be?
6. What happens to the local square-grid picture as r gets smaller?

Note

The local square-grid visualization does not produce one matrix for the whole nonlinear map. It shows one matrix approximation near one base point.

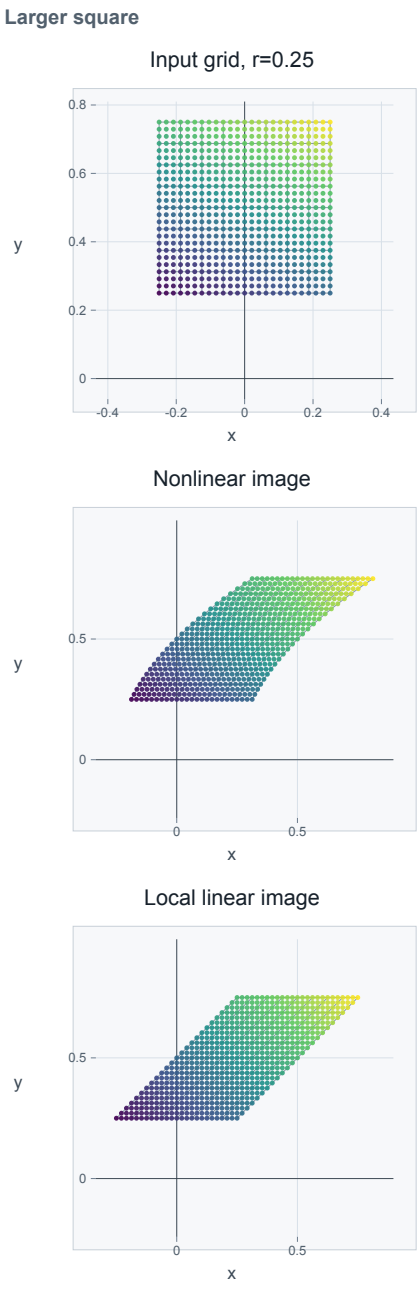


Figure. Local square-grid visualization near $\mathbf{a} = \begin{bmatrix} 0 \\ 1/2 \end{bmatrix}$ for the larger input square $r = 0.25$. The three panels show the input grid, the nonlinear image, and the local linear image on ticked axes.

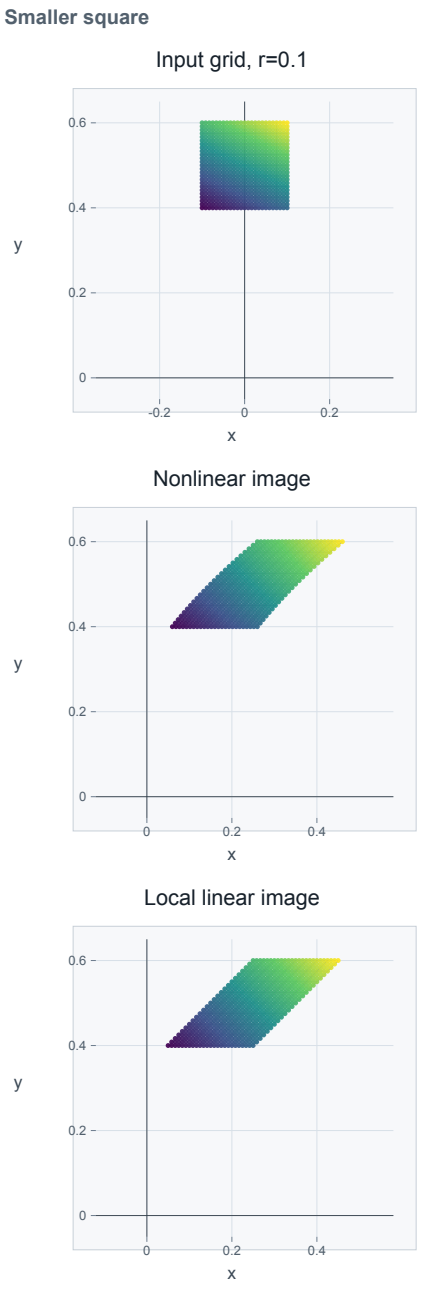


Figure. Local square-grid visualization near $\mathbf{a} = \begin{bmatrix} 0 \\ 1/2 \end{bmatrix}$ for the smaller input square $r = 0.1$. The nonlinear image is closer to the local linear image than in the larger-square visualization.

Activity: Reading a local linear prediction in SymPy (U3-LO5)

Use SymPy to revisit the nonlinear shear $F(x, y) = (x + y^2, y)$ from Local square-grid visualization (U3-LO5). We evaluate its Jacobian at $\mathbf{a} = (0, 1/2)$, then use that matrix to predict the output after a

Activity: Reading a local linear prediction in SymPy (U3-LO5) (continued)

small input change. The command `sp.Rational(p, q)` represents the fraction p/q exactly. As in Using a SymPy null-space vector (U2-LO3, U2-LO4, U2-LO6), `*` between SymPy matrices means matrix multiplication.

```
import sympy as sp

x, y = sp.symbols("x y")
F = sp.Matrix([x + y**2, y])
a = sp.Matrix([0, sp.Rational(1, 2)])
h = sp.Matrix([sp.Rational(1, 10), -sp.Rational(1, 10)])

J = F.jacobian([x, y])
Ja = J.subs({x: a[0], y: a[1]})
Fa = F.subs({x: a[0], y: a[1]})
prediction = Fa + Ja*h
actual = F.subs({x: a[0] + h[0], y: a[1] + h[1]})

print(Ja)
print(Ja*h)
print(prediction)
print(actual)
print(actual - prediction)
```

Output:

```
Matrix([[1, 1], [0, 1]])
Matrix([[0], [-1/10]])
Matrix([[1/4], [2/5]])
Matrix([[13/50], [2/5]])
Matrix([[1/100], [0]])
```

1. What are the shapes of `Ja`, `h`, and `Ja*h`? Identify the shear matrix from Horizontal shear (U1-LO4). Why does `Ja*h` represent a predicted change rather than the predicted output?

Activity: Reading a local linear prediction in SymPy (U3-LO5) (continued)

2. Write the matrix calculation represented by `Fa + Ja*h`. Which output coordinate is predicted exactly for this input change, and what is the error in the other coordinate?

3. If `h` is replaced by `h/2` before computing `prediction` and `actual`, predict the new error. Explain the change using the error formula in Local square-grid visualization (U3-LO5).

Activity: Interpreting a Jacobian's null space in SymPy (U3-LO5, U2-LO3)

In Using a SymPy null-space vector (U2-LO3, U2-LO4, U2-LO6), a null-space vector described an input change that a linear map forgets. Apply the same tools to the local linear map of $F(x, y) = (x + y, x^2 + y^2)$ at $\mathbf{a} = (1, 1)$. Recall that `.rref()` returns the reduced matrix and zero-based pivot-column indices, while `.nullspace()` returns a list of basis column matrices.

```
import sympy as sp

x, y = sp.symbols("x y")
F = sp.Matrix([x + y, x**2 + y**2])
J = F.jacobian([x, y])
Ja = J.subs({x: 1, y: 1})

R, pivots = Ja.rref()
basis = Ja.nullspace()
v = basis[0]

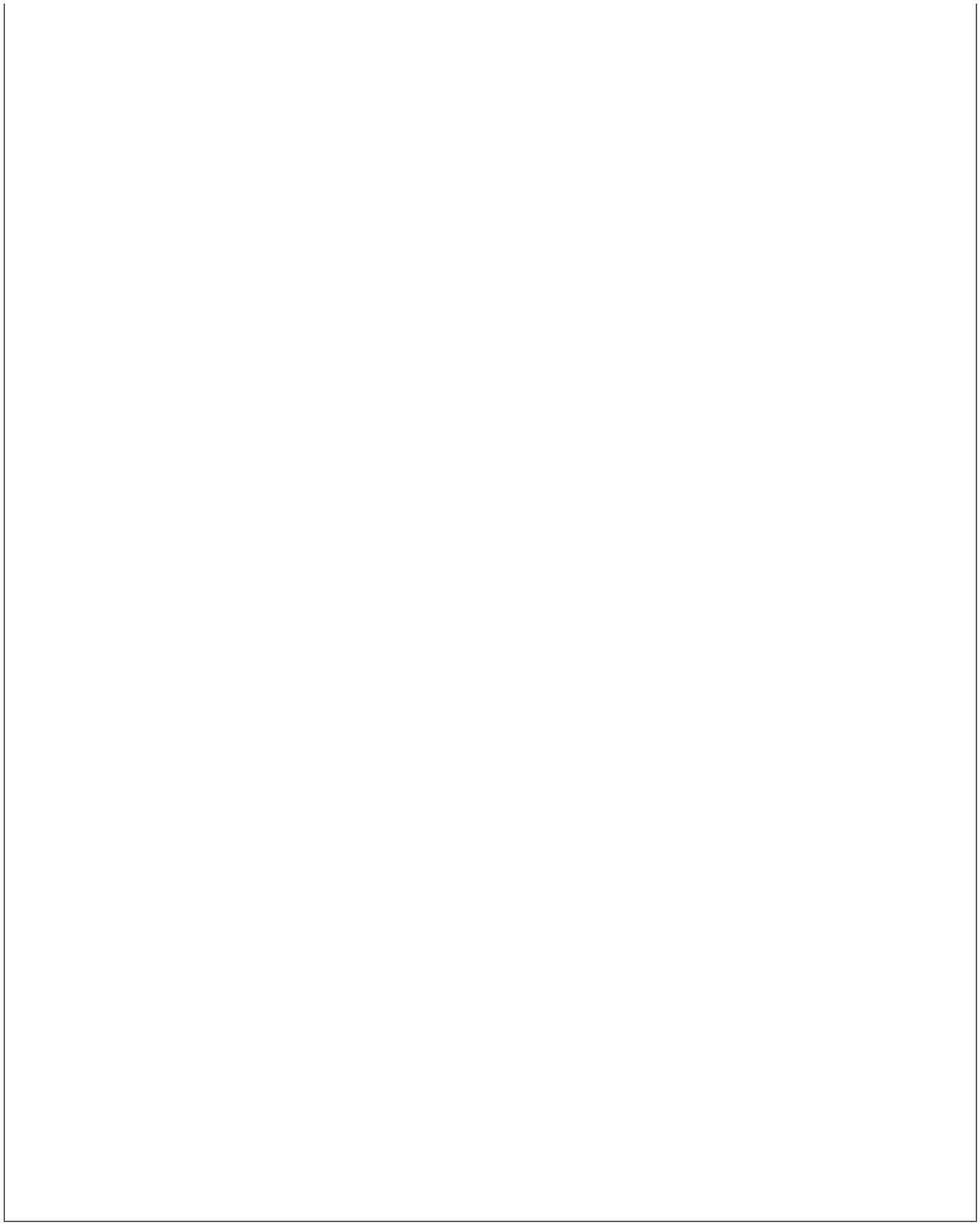
print(Ja)
print(R, pivots)
print(len(basis), v)
print(Ja*v)
```

Output:

Activity: Interpreting a Jacobian's null space in SymPy (U3-LO5, U2-LO3) (continued)

```
Matrix([[1, 1], [2, 2]])
Matrix([[1, 1], [0, 0]]) (0,)
1 Matrix([[-1], [1]])
Matrix([[0], [0]])
```

1. What are the rank and nullity of `Ja`? Using its original pivot column, describe the possible first-order output changes $J_F(\mathbf{a})\mathbf{h}$.
2. Let $\mathbf{v} = (-1, 1)$. What does the local linear approximation predict for $F(\mathbf{a} + t\mathbf{v})$ when t is small?
3. Substitute $\mathbf{a} + t\mathbf{v} = (1 - t, 1 + t)$ directly into F. Does the zero product `Ja*v` imply that the nonlinear output stays exactly constant? Find the actual change when $t = 1/10$.
4. Predict `J.subs({x: 1, y: 0}).nullspace()`. What does this tell you about how the local linear map depends on the base point?



The approximation in this unit is local. Near one input $\mathbf{a}$, a differentiable function behaves like

$$F(\mathbf{a}+\mathbf{h}) \approx F(\mathbf{a}) + J_F(\mathbf{a})\mathbf{h}.$$

Unit 4 studies a different kind of approximation:

$$\min_x \|Ax - b\|.$$

That is a global best approximation by a vector in a subspace. Unit 3 asks what a nonlinear function looks like near one point; Unit 4 asks which allowable vector is closest overall.